

1. Various Types of AI

AI is generally categorized based on two main criteria: **Capabilities** and **Functionalities**.

- **Based on Capabilities:**

- **Narrow AI (Weak AI):** Specialized in one task (e.g., Siri, Alexa, or recommendation systems).
- **General AI (Strong AI):** Theoretical AI that can perform any intellectual task a human can.
- **Super AI:** A hypothetical stage where AI surpasses human intelligence across all fields.

- **Based on Functionalities:**

- **Reactive Machines:** No memory; they only respond to current scenarios (e.g., Deep Blue chess computer).
- **Limited Memory:** Can use past data to make better decisions (e.g., Self-driving cars).
- **Theory of Mind:** Theoretical AI that understands human emotions and beliefs.
- **Self-Awareness:** The ultimate stage where AI has its own consciousness and self-awareness.

2. Differentiate between Supervised and Unsupervised Learning

Feature	Supervised Learning	Unsupervised Learning
1. Definition	A type of machine learning where the model is trained on a labeled dataset	A type of machine learning where the model identifies hidden

	(input-output pairs) to predict future outcomes.	patterns or structures within unlabeled data.
2. Goal	To map an input (\$x\$) to an output (\$y\$) based on provided examples.	To find the underlying structure or distribution in the data.
3. Supervision	Operates under the guidance of a "teacher" or labeled data.	Operates on its own to discover information.
4. Types	Categorized into Regression and Classification.	Categorized into Clustering and Association.
5. Accuracy	Highly accurate as it uses verified labels for training.	Less accurate; results are subjective and depend on pattern discovery.
6. Examples	Spam detection (Gmail), House price prediction.	Customer segmentation (Netflix), Image compression.

3. Informed vs. Uninformed Search Algorithms

Feature	Uninformed Search (Blind Search)	Informed Search (Heuristic Search)
1. Definition	A search strategy that explores the search space without any domain knowledge or information about the distance to the goal.	A search strategy that uses heuristic functions (additional information) to estimate the cost/distance to the goal.
2. Knowledge	Knows only the start state, the goal state, and valid operators.	Uses domain-specific knowledge to guide the search towards the goal efficiently.
3. Performance	Generally slower and consumes more memory (explores all directions).	Faster and more memory-efficient as it focuses on the most promising paths.
4. Heuristic ($h(n)$)	Does not use a heuristic function.	Uses a heuristic function $h(n)$ to rank nodes.
5. Complexity	High time and space complexity.	Lower complexity due to targeted exploration.
6. Examples	Breadth-First Search (BFS),	A* Search, Greedy Best-First

	Depth-First Search (DFS).	Search.
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4. Application Areas of AI

AI is revolutionizing multiple industries:

- **Healthcare:** Medical imaging analysis, drug discovery, and robotic surgery.
- **Finance:** Fraud detection, algorithmic trading, and credit scoring.
- **E-commerce:** Personalized recommendations (Amazon/Netflix) and chatbots.
- **Automotive:** Self-driving cars and traffic management systems.
- **Education:** Personalized learning paths and automated grading.
- **Gaming:** Creating intelligent Non-Player Characters (NPCs).

5. PEAS Description for a House-Cleaning Robot

PEAS stands for **P**erformance, **E**nvironment, **A**ctuators, and **S**ensors. It is used to define the task environment of an intelligent agent.

For a House-Cleaning Robot:

- **P (Performance Measure):** Cleanliness of the floor, time taken, battery efficiency, safety (not hitting pets/objects).
 - **E (Environment):** Rooms, hallways, carpets, furniture, stairs, and inhabitants (humans/pets).
 - **A (Actuators):** Wheels or tracks for movement, brushes for sweeping, vacuum motor, water spray for mopping.
 - **S (Sensors):** Camera, Lidar/Ultrasonic sensors (for obstacles), Cliff sensors (to avoid stairs), Bump sensors, Dust sensors.
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6. Problems Robots Can Solve

Robots are primarily used to solve problems categorized as the "4 Ds":

- **Dirty:** Tasks like sewage cleaning, house cleaning, or waste management.
 - **Dull:** Repetitive assembly line tasks in manufacturing (e.g., picking and placing).
 - **Dangerous:** Defusing bombs, deep-sea exploration, or space missions.
 - **Difficult:** High-precision tasks like microsurgery or heavy lifting in logistics.
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7. Components of an AI Program

An AI program typically consists of several core technical components:

1. **Input/Perception:** Gathering data through sensors or digital inputs (Data Collection).
2. **Knowledge Base:** A repository of facts, rules, and data that the AI uses to understand the domain.
3. **Inference Engine:** The "brain" that applies logical rules to the knowledge base to derive new information or make decisions.
4. **Learning Component:** Algorithms that allow the system to improve its performance over time based on experience.
5. **Output/Actuation:** The final action taken, whether it's a digital response (chatbot) or physical movement (robot).

In AI, the **PEAS description** is a framework used to specify the characteristics of an intelligent agent and its task environment. It is the first step in designing any rational agent.

8. Describe PEAS description . Illustrate the PEAS description for Vacuum Cleaner Agent.

Breakdown of PEAS

- **Performance Measure:** The criteria that determine how successful the agent is. It defines what "success" looks like.
 - **Environment :** The external surroundings or world in which the agent operates.
 - **Actuators :** The "parts" that allow the agent to act upon the environment or perform tasks.
 - **Sensors:** The "inputs" that allow the agent to perceive its environment and gather data.
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2. PEAS Description for a Vacuum Cleaner Agent

For an automated vacuum cleaner (like a Roomba), the PEAS description identifies its goals, its working area, and how it interacts with its surroundings.

Component	Description for Vacuum Cleaner Agent
Performance	Cleanliness (percentage of dirt removed), time taken, battery consumption, and safety (not falling or breaking items).
Environment	Wooden floors, carpets, obstacles (furniture, shoes), stairs, pets, and walls.
Actuators	Motorized wheels for movement, rotating brushes for sweeping, vacuum motor for suction, and speakers/lights for status.
Sensors	Camera/LiDAR for mapping, Dirt sensors to detect debris, Bump sensors for obstacles, and Cliff sensors to avoid stairs.

9. Classify the types of chaining.

Illustrate forward chaining and backward chaining algorithms with suitable examples.

10.Explain the PEAS description for Automated taxi driver.

- **Performance (P):** The goals or "success criteria"—**Safety** (avoiding crashes), **Speed** (fastest route), **Legality** (obeying traffic laws), and **Passenger Comfort** (smooth ride).
- **Environment (E):** The external surroundings—**Roads** (highways/streets), **Traffic** (cars/buses), **Pedestrians**, **Weather** (rain/fog), and **Street Signs**.
- **Actuators (A):** The parts used to act/move—**Steering Wheel**, **Accelerator**, **Brakes**, **Signal Lights**, and **Horn**.
- **Sensors (S):** The parts used to perceive—**Cameras** (traffic lights/lanes), **LiDAR/Radar**

(distance to objects), **GPS** (location), and **Speedometer**.

Knowledge-based agents are intelligent agents that reason about their world using a formal internal representation of knowledge. According to **GeeksforGeeks** and standard AI theory (Russell & Norvig), there are three distinct levels at which we can describe these agents.

11. Describe levels of knowledge based agents.

The Three Levels of Knowledge-Based Agents

Level	Description	Key Focus
1. Knowledge Level	The highest and most abstract level. It describes what the agent knows and what its goals are, regardless of how the knowledge is implemented.	Goals and Facts
2. Logical Level	The level where knowledge is encoded into formal sentences or logic (such as First-Order Logic). It defines how the facts are structured.	Representation
3. Implementation Level	The lowest level, focusing on the actual physical representation or data structures (like linked lists or arrays) inside the computer system.	Data Structures

Detailed Explanation

1. Knowledge Level (or Linguistic Level)

At this level, we specify the agent's behavior by describing what it knows. We don't care about the computer code; we only care about the information.

- **Example:** "The agent knows that if it is raining, the grass will be wet, and its goal is to keep the floor dry."

2. Logical Level (or Functional Level)

This level bridges the gap between abstract knowledge and computer code. Here, the knowledge is translated into logical notations. The agent uses an **Inference Engine** to manipulate these logical sentences to derive new information.

- **Example:** $IsRaining(x) \rightarrow GrassIsWet(x)$

3. Implementation Level (or Physical Level)

This is the architecture where the knowledge actually "lives" in the machine. It involves the specific programming language (Python, LISP, Prolog) and the data structures (symbols, hashes, or pointers) used to store the Knowledge Base (KB).

- **Example:** Using a Python dictionary or a SQL database to store facts and rules.
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